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Comparative Review of Credit Card Fraud Detection using Machine Learning and Concept Drift Techniques

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Abstract:
Credit card fraud is a significant concern for financial institutions and cardholders alike. As fraudulent activities become more sophisticated, traditional rule-based approaches struggle to keep up. This has led to the adoption of machine learning techniques for fraud detection, which have shown promising results. However, the dynamic nature of credit card fraud poses a challenge due to the concept drift phenomenon. Concept drift refers to the changes in the underlying data distribution over time, requiring models to adapt and evolve to maintain their effectiveness. This research paper aims to provide a comprehensive comparative review of credit card fraud detection methods using machine learning and concept drift techniques. This literature review provides an overview of relevant studies comparing credit card fraud detection using machine learning techniques and concept drift handling methods. The paper evaluates the performance, strengths, and limitations of various approaches in addressing credit card fraud detection under concept drift scenarios.

1.0 Introduction

Credit card fraud detection is a critical task for financial institutions due to the increasing sophistication of fraudulent activities. Traditional rule-based approaches often struggle to keep up with evolving fraud patterns, leading to the adoption of machine learning techniques. However, the dynamic nature of credit card fraud poses a challenge, as the underlying data distribution can change over time, requiring models to adapt to concept drift. This literature review provides an overview of relevant studies comparing credit card fraud detection using machine learning techniques and concept drift handling methods.

Machine Learning Techniques for Credit Card Fraud Detection:

In this study, Bhattacharyya *et al*. (2019) evaluated different machine learning algorithms, including logistic regression, decision trees, random forests, and support vector machines, for credit card fraud detection. The authors concluded that ensemble methods, such as random forests, outperformed other techniques in terms of accuracy and efficiency. A comparative study conducted by Ahmed *et al*. (2020) assessed the performance of various machine learning models, including logistic regression, k-nearest neighbours, and artificial neural networks, for credit card fraud detection. The study revealed that neural networks achieved the highest detection accuracy and were more effective in handling complex fraud patterns.

In another research work, Liu *et al*. (2021) compared the performance of different machine learning algorithms, including gradient boosting machines, support vector machines, and deep learning models, for credit card fraud detection. The results showed that deep learning models, particularly convolutional neural networks, demonstrated superior performance in detecting fraudulent transactions.

In the study on concept drift detection, Bifet *et al*. (2019) proposed the Hoeffding Adaptive Trees (HAT) algorithm, which dynamically adapts to changing data distributions. The HAT algorithm showed promising results in handling concept drift for credit card fraud detection, outperforming traditional batch learning approaches. Nguyen *et al*. (2020) proposed a novel concept drift handling method called Adaptive Random Forest (ARF) for credit card fraud detection. ARF combines ensemble learning with incremental learning, allowing the model to adapt to evolving fraud patterns. The study demonstrated that ARF effectively handled concept drift and maintained high detection accuracy. Liu *et al*. (2021) proposed a hybrid approach that combined online learning and change detection methods for credit card fraud detection. The approach continuously updated the model using incoming data and detected concept drift using statistical change detection techniques. The hybrid approach achieved better performance compared to traditional batch learning methods.

1.1 Problem Statement

The current challenge of Credit Card Fraud Detection using Machine Learning and Concept Drift Techniques is to address the challenge of effectively detecting credit card fraud using machine learning algorithms in the presence of concept drift. Concept drift refers to the phenomenon where the underlying data distribution of fraudulent transactions changes over time, requiring models to adapt and evolve to maintain their effectiveness. The dynamic nature of credit card fraud patterns poses a challenge for traditional rule-based approaches, leading to the adoption of machine learning techniques. However, the effectiveness of these techniques can be compromised by concept drift, as models trained on historical data may not accurately capture evolving fraud patterns. Therefore, the problem involves developing and evaluating machine learning models and concept drift handling techniques that can accurately detect credit card fraud in the face of changing data distributions and dynamic fraud patterns. The goal is to achieve high detection accuracy, minimize false positives, and provide robust fraud detection capabilities in real-time or near-real-time scenarios.

This literature review provides an overview of relevant studies comparing credit card fraud detection using machine learning techniques which are not limited to only concept drift handling methods. It evaluates the performance, strengths, and limitations of various approaches in addressing credit card fraud detection under concept drift scenarios and also offers future areas of research.

2.0 Credit Card Fraud Detection

2.1 Overview of Credit Card Fraud

Credit card fraud is a type of financial fraud that involves the unauthorized use of another person's credit card information to make fraudulent transactions or gain unauthorized access to funds. It is a widespread and significant problem that affects individuals, businesses, and financial institutions worldwide. Credit card fraud can occur through various means, including physical theft, online scams, data breaches, and card skimming devices.

The impact of credit card fraud is substantial, leading to financial losses for individuals, businesses, and financial institutions. Fraudulent transactions can result in unauthorized charges, identity theft, and compromised personal and financial information. The consequences of credit card fraud extend beyond financial losses, as victims often experience stress, inconvenience, and the need to go through lengthy procedures to resolve fraudulent activities.

To combat credit card fraud, financial institutions and card issuers employ various preventive measures and detection techniques. These measures include:

i. Fraud Monitoring Systems: Financial institutions implement advanced fraud monitoring systems that analyze transaction data in real-time, looking for patterns or anomalies indicative of fraudulent activity. These systems use rule-based algorithms, anomaly detection techniques, and machine learning models to identify suspicious transactions and trigger alerts for further investigation.

ii. Two-Factor Authentication: Many credit card issuers have implemented additional layers of security, such as two-factor authentication. This involves requiring users to provide additional verification, such as a one-time password or biometric authentication, to authorize a transaction.

iii. EMV Chip Technology: EMV (Europay, Mastercard, and Visa) chip technology has been widely adopted to replace traditional magnetic stripe cards. EMV cards store encrypted transaction data and generate unique transaction codes for each purchase, making it more difficult for fraudsters to clone or counterfeit cards.

iv. Tokenization: Tokenization is a technique that replaces sensitive cardholder data with unique tokens. When a transaction occurs, the token is used instead of the actual card information, reducing the risk of exposure to fraudsters in case of a data breach.

v. Machine Learning-Based Fraud Detection: Machine learning algorithms are increasingly being used to detect credit card fraud. These algorithms analyze large volumes of transaction data, identify patterns, and learn from historical fraudulent activities to detect new and emerging fraud patterns. Machine learning models can continuously adapt and improve their detection capabilities as new data becomes available.

Despite these preventive measures and detection techniques, credit card fraud remains a persistent and evolving problem. Fraudsters continually develop new tactics and techniques to circumvent security measures, requiring constant vigilance and innovation in fraud detection and prevention strategies. The ongoing challenge is to stay one step ahead of fraudsters by leveraging advanced technologies and constantly evolving fraud detection methods.

2.2 Importance of Fraud Detection

Fraud detection is of paramount importance for individuals, businesses, and financial institutions. Here are several key reasons highlighting the significance of fraud detection:
Fraud detection plays a crucial role in preventing financial losses. Detecting fraudulent activities early allows for prompt action, minimizing the impact of unauthorized transactions and reducing the financial burden on individuals, businesses, and financial institutions. Fraud incidents can erode customer trust in financial institutions and businesses. Implementing robust fraud detection measures demonstrate a commitment to customer security, fostering trust and loyalty among customers. Successful fraud detection helps mitigate the risk of reputation damage for businesses and financial institutions. By promptly identifying and addressing fraudulent activities, organizations can protect their brand image and maintain a positive reputation in the market.

Many industries are subject to strict regulations regarding fraud prevention and detection. Adhering to these regulations is not only a legal requirement but also ensures ethical business practices and safeguards the interests of customers and stakeholders. Fraud detection aids in preventing identity theft, a serious crime that can have far-reaching consequences for individuals. By detecting and stopping fraudulent activities, organizations can help protect the personal and financial information of their customers, reducing the risk of identity theft. Implementing effective fraud detection measures can lead to significant cost savings for businesses and financial institutions. By reducing the occurrence and impact of fraud, organizations can save on operational costs associated with investigation, reimbursement, and legal proceedings. Fraudulent activities can have a destabilizing effect on financial systems. By detecting and preventing fraud, financial institutions contribute to maintaining the overall stability and integrity of the financial system.

Fraud, especially large-scale fraud operations, is often linked to organized crime syndicates. Detecting and disrupting these criminal networks through effective fraud detection measures contributes to overall crime prevention and public safety. Fraud detection systems can serve as an early warning system, alerting organizations to emerging fraud trends and evolving fraud techniques. This allows organizations to proactively update their fraud prevention strategies and stay ahead of fraudsters. Fraud detection systems often involve robust data security measures to protect sensitive customer information. By implementing strong security practices and encryption techniques, organizations enhance data protection and minimize the risk of data breaches.

Fraud detection is essential for financial security, customer trust, regulatory compliance, and maintaining the integrity of financial systems. By investing in effective fraud detection measures, organizations can safeguard their finances, reputation, and the interests of their customers and stakeholders.

2.3 Traditional Rule-Based Approaches

Traditional rule-based approaches have long been used for credit card fraud detection. These approaches involve defining a set of rules or heuristics based on known fraud patterns and indicators to identify potentially fraudulent transactions. Research on Credit Card Fraud Detection using Rule-based, and Decision Tree Algorithm was implemented (Zhao *et al*, 2012; Yates, 2003). A review of traditional rule-based approaches in credit card fraud detection was carried out in (Chandola *et al*, 2009; Srinivasan, *et al*, 2019; Bhattacharyya, *et al*, 2010). Traditional rule-based methods leverage predefined rules and heuristics to identify fraudulent transactions based on specific patterns or conditions. A comparative study on traditional rule-based techniques in (Alazab *et al,* 2015; Marsono and Sundaraswaran, 1994; Elkhani and Yusof, 2013) highlights key studies, techniques, and their performance in credit card fraud detection. Additionally, it discusses the limitations of these approaches and suggests potential directions for future research in the field.

Here's an overview of credit card fraud detection using traditional rule-based approaches:

- Rule-Based Fraud Detection Systems: Rule-based systems employ a predefined set of rules that flag transactions based on specific criteria. These rules are typically designed based on historical fraud patterns and expert knowledge. For example, rules may include conditions such as high-value transactions, unusual purchase locations, multiple transactions within a short period, or transactions outside the cardholder's usual spending habits.
- Threshold-Based Approaches: Threshold-based approaches define thresholds for specific transaction attributes, such as transaction amount, transaction frequency, or geographic location. Transactions that exceed these predefined thresholds are flagged as potentially fraudulent. For example, if a transaction exceeds a certain dollar amount or occurs in a country the cardholder has never visited before, it may trigger an alert.
- Velocity Checks: Velocity checks analyze the rate or frequency of transactions to identify suspicious patterns. For instance, if multiple transactions from different merchants occur within a short time frame, it may indicate fraudulent activity. Velocity checks aim to identify unusual transaction volumes that deviate from the cardholder's typical spending behaviour.
- Address Verification System (AVS): AVS is a rule-based system that verifies the billing address provided during a transaction against the address on file with the card issuer. If there is a mismatch or incomplete address information, the transaction may be flagged for further investigation.
- Card Verification Code (CVC) Checks: CVC checks involve validating the three-digit security code (CVC/CVV) provided during a transaction. If the CVC does not match the cardholder's information, it raises suspicion of potential fraud.
- Country and Merchant Risk Scores: Some rule-based systems assign risk scores to different countries or merchants based on historical fraud patterns. Transactions originating from high-risk countries or involving high-risk merchants may trigger additional scrutiny or require additional verification steps.

Advantages of Traditional Rule-Based Approaches:

- Transparency: Rule-based approaches are transparent, as the rules and criteria used for fraud detection are explicitly defined and understandable.
- Quick Implementation: Rule-based systems can be implemented relatively quickly, as they rely on predefined rules and criteria.
- Low False Positive Rates: Rule-based approaches often have lower false positive rates, as the rules are designed to target specific fraud patterns.

Limitations of Traditional Rule-Based Approaches:

- Limited Adaptability: Traditional rule-based systems may struggle to adapt to new or evolving fraud patterns. They heavily rely on predefined rules and may not effectively capture emerging fraud schemes.
- High False Negative Rates: Rule-based systems may miss sophisticated fraud patterns that do not conform to predefined rules, leading to false negatives.

- Limited Scalability: As fraud patterns become more complex, managing and updating a large number of rules can become cumbersome and impractical.
- Inability to Detect Unknown Fraud Patterns: Rule-based approaches are effective in detecting known fraud patterns but are less capable of identifying new or unknown fraud patterns.

Overall, while traditional rule-based approaches have been widely used in credit card fraud detection, their limitations in adapting to evolving fraud patterns have led to the exploration of more advanced techniques such as machine learning and concept drift handling methods.

3.0 Machine Learning Techniques for Fraud Detection

Fraud detection is a critical task in various domains, including finance, insurance, and e-commerce. With the increasing complexity and sophistication of fraudulent activities, traditional rule-based approaches struggle to keep pace with emerging fraud patterns. Machine learning techniques have emerged as powerful tools for fraud detection, leveraging their ability to learn from large volumes of data and identify complex patterns indicative of fraudulent behaviour. Fraud detection is a challenging task that requires timely and accurate identification of fraudulent activities while minimizing false positives. Machine learning techniques have gained prominence due to their ability to analyze large datasets, detect intricate patterns, and adapt to evolving fraud schemes. Machine learning techniques have proven useful in solving the credit card fraud detection problem. Logistic regression, decision trees, random forests, SVMs, neural networks, ensemble techniques, anomaly detection, and concept drift handling methods are commonly employed in fraud detection applications. Each technique has its strengths and limitations, and the choice of algorithm depends on the specific requirements and characteristics of the fraud detection task. Some of the techniques used in credit card fraud detection are described in the next section.

Most literatures have explored the application of machine learning techniques for fraud detection (Bhusari *et al,* 2016; Phua *et al*, 2004; Jøsang, 2006). Machine learning methods leverage data-driven algorithms to automatically learn patterns and detect fraudulent activities in various domains. The studies highlight key studies, techniques, and their performance in fraud detection using machine learning. It discusses the advantages and challenges of machine learning approaches and suggests potential directions for future research in the field.

3.1 Supervised Learning

Some literatures examine the application of supervised learning techniques in credit card fraud detection (Bahmani-Firouzi and Ghonoodi, 2020; Kshirasagar and Dixit, 2019; Idowu *et al*, 2017; Chen and Varshavsky, 2004; Aydin and Cambazoglu, 2020). Supervised learning methods utilize labelled data to train predictive models that can classify transactions as either fraudulent or legitimate. These studies explore various algorithms, feature engineering approaches, evaluation metrics, and challenges associated with the application of supervised learning in this domain.

This section presents a detailed analysis of various supervised machine learning algorithms used in fraud detection. Key algorithms and techniques include:

1. Logistic Regression: Logistic regression is a widely used classification algorithm in fraud detection. Its simplicity, interpretability, and efficiency make it suitable for detecting straightforward fraud patterns. However, logistic regression may struggle with capturing complex and nonlinear fraud schemes.

2. Decision Trees and Random Forests: Decision trees offer an intuitive representation of rules for fraud detection. They can handle both categorical and numerical features and provide interpretability. Random forests, an ensemble technique of decision trees, improve classification accuracy and generalize well to detect complex fraud patterns. Random Forests offer a robust and accurate approach for credit card fraud detection. By leveraging the ensemble of decision trees, Random Forests can effectively handle imbalanced data, capture non-linear fraud patterns, provide insights into feature importance, and offer reasonable interpretability.
3. Support Vector Machines (SVM): SVMs are effective in separating fraudulent transactions from genuine ones by mapping data into a higher-dimensional space. They can handle high-dimensional data and capture nonlinear relationships, enabling the detection of intricate fraud patterns. However, SVMs may face scalability challenges with large datasets.
4. Neural Networks: Neural networks, particularly deep learning models, have shown promising results in fraud detection. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) can automatically extract features and capture intricate patterns, leading to improved detection accuracy. However, deep learning models require substantial computational resources and extensive data for training.
5. k-Nearest Neighbors (k-NN): It is a non-parametric and instance-based algorithm that classifies new instances based on the majority vote of their k nearest neighbours in the feature space. k-NN has its advantages, such as adaptability and interpretability, it also has limitations. It can be sensitive to noise and outliers, and its performance may degrade in high-dimensional spaces. However, when properly applied and optimized, k-NN can be a valuable tool for credit card fraud detection, especially in scenarios where interpretability and adaptability are crucial.
6. Naïve Bayes: Despite its simplicity, Naive Bayes has proven to be effective in many classification tasks, including fraud detection. It is based on Bayes' theorem, which calculates the probability of an event based on prior knowledge. While it may not capture complex dependencies between features, Naive Bayes can provide a baseline performance and serve as a valuable addition to the ensemble of other machine learning techniques used in fraud detection systems.

3.2 Unsupervised Learning

Unsupervised machine learning techniques can be useful in credit card fraud detection for identifying patterns and anomalies in the data. While most fraud detection approaches utilize supervised techniques, unsupervised techniques can be applied for exploratory analysis or as part of a hybrid approach. Here is a list of unsupervised machine learning techniques applied in credit card fraud detection:

1. Clustering:

a. K-means clustering: Grouping similar transactions together based on their features.
b. Hierarchical clustering: Creating a hierarchy of clusters to identify different levels of fraud patterns.

2. Anomaly Detection:

a. Isolation Forest: Detecting anomalies by isolating instances in a tree-based structure.
b. Local Outlier Factor (LOF): Identifying local deviations from the overall pattern.
c. One-Class SVM (Support Vector Machine): Learning the boundaries of normal data and identifying deviations.

3. Autoencoders:
a. Variational Autoencoders (VAEs): Unsupervised neural network models that learn the underlying distribution of the data, useful for detecting outliers and anomalies.
4. Self-Organizing Maps (SOM):
a. Neural network-based models that create a low-dimensional representation of the data and reveal clusters or patterns.
5. Association Rule Mining:
a. Apriori algorithm: Discovering associations and correlations between different features or transactions.
6. Density-Based Techniques:
a. DBSCAN (Density-Based Spatial Clustering of Applications with Noise): Identifying dense regions of data points, useful for detecting fraud clusters.
7. Outlier Detection:
a. Elliptic Envelope: Detecting outliers by assuming the data follows an elliptical distribution.

These unsupervised techniques can help in identifying unusual patterns, clusters, or anomalies within credit card transaction data. They can be utilized to perform exploratory analysis, identify potentially fraudulent activities, or pre-process data before applying supervised techniques. However, it's worthy of note that unsupervised techniques may have limitations in terms of interpretability and may require further investigation or validation before taking any action in real-world fraud detection scenarios.

3.3 Semi-Supervised Learning

Semi-supervised machine learning techniques are useful in credit card fraud detection when there is a limited amount of labelled fraud data available. These techniques leverage both labelled and unlabelled data to improve the detection performance. Here is a list of semi-supervised machine learning techniques that are applied in solving credit card fraud detection problem.

1. Self-Training: Self-training involves using a small set of labelled fraud data to train a classifier and then applying the classifier to unlabelled data. The most confident predictions on the unlabelled data are added to the labelled data, and the classifier is retrained iteratively.
1. Co-training: Co-training utilizes multiple views or feature sets of the data. Initially, classifiers are trained on different subsets of features. The classifiers then label unlabelled instances, and instances with consistent predictions from both classifiers are added to the labelled data for retraining.
2. Multi-view Learning: Multi-view learning combines information from multiple sources or views of the data to improve fraud detection performance. Each view represents a different perspective or representation of the data, which can be beneficial in capturing diverse patterns.
3. Transductive Learning: Transductive learning focuses on leveraging the unlabelled data in the same domain as the labelled data to improve classification performance. It uses the structure or distribution of the unlabelled data to make predictions on new instances.
4. Expectation-Maximization (EM) Algorithm: The EM algorithm is an iterative method that estimates the parameters of a model by maximizing the likelihood function. In credit card fraud detection, it can be used to estimate the parameters of a mixture model, such as Gaussian Mixture Models (GMM), by incorporating both labelled and unlabelled data.

5. Label Propagation: Label propagation is a technique that assigns labels to unlabelled instances based on the labels of their neighbouring instances. It propagates labels through the data graph, leveraging the local structure of the data to make predictions.
   These semi-supervised techniques aim to leverage the available labelled fraud data while effectively utilizing the information present in the larger set of unlabelled data. By incorporating unlabelled data, these techniques can improve the performance of credit card fraud detection systems, especially in scenarios where labelled fraud data is limited or costly to obtain.

3.4 Ensemble Methods

Ensemble machine learning techniques combine the predictions of multiple individual models to make more accurate and robust predictions. These techniques have been widely used in credit card fraud detection to improve the overall detection performance (Teo *et al*, 2013; García *et al*, 2012; Ruta *et al*, 2013; Brown and Kuncheva, 2013). Here is a list of ensemble machine learning techniques commonly utilized in credit card fraud detection:

1. Random Forest: Random Forest is an ensemble method that constructs multiple decision trees using random subsets of the training data and features. The final prediction is made by aggregating the predictions of all individual trees.
2. Gradient Boosting: Gradient Boosting is an ensemble technique that builds multiple weak models (e.g., decision trees) sequentially, with each model learning from the mistakes of its predecessors. Examples include AdaBoost, XGBoost, and LightGBM.
3. Bagging: Bagging (Bootstrap Aggregating) involves training multiple models on different bootstrapped samples of the training data. The final prediction is typically made by averaging or voting across the predictions of all models. Examples include Bagging Classifier and Random Subspace.
4. Stacking: Stacking combines the predictions of multiple models by training a meta-model on the outputs of individual models. The individual models act as base models, and their predictions serve as input features for the meta-model.
5. Voting: Voting ensembles combine the predictions of multiple models by majority voting or weighted voting. It can be done with different algorithms or by varying hyperparameters to improve overall performance.
6. Adversarial Training: Adversarial training involves training a model on both the original data and adversarial samples generated to mimic potential fraud attempts. This technique helps improve the model's robustness to adversarial attacks.
7. Rotation Forest: Rotation Forest applies a dimensionality reduction technique, such as Principal Component Analysis (PCA), to each base model in an ensemble. It aims to enhance diversity among the individual models by transforming the input features.
8. Boosting: Boosting algorithms, such as AdaBoost and Gradient Boosting, sequentially train multiple weak models, with each model focusing on the previously misclassified instances. The final prediction is made by combining the predictions of all weak models. AdaBoost combines multiple weak classifiers to create a strong classifier and iteratively trains classifiers on different subsets of the data, assigning higher weights to misclassified instances to focus on the difficult cases. AdaBoost can be sensitive to concept drift as the distribution of the data may change between training iterations. Techniques such as periodically retraining the AdaBoost model or using adaptive learning algorithms can help address concept drift and maintain the effectiveness of the classifier over time.

9. Neural Network Ensembles: Neural network ensembles combine predictions from multiple neural network models, either with different architectures, initializations, or trained on different subsets of the data. The ensemble approach helps improve generalization and capture diverse patterns.

These ensemble machine learning techniques aim to combine the strengths of multiple models to improve credit card fraud detection performance. By leveraging the diversity and collective knowledge of individual models, ensemble techniques can enhance accuracy, robustness, and generalization capabilities in detecting fraudulent activities.

4.0 Concept Drift in Credit Card Fraud Detection

Concept drift refers to the phenomenon where the underlying data distribution in a problem domain changes over time. In the context of credit card fraud detection, concept drift can occur when the patterns and characteristics of fraudulent transactions evolve or when new types of fraud emerge. It poses a significant challenge for fraud detection systems as models trained on historical data may become less effective in detecting new or evolving fraud patterns.

Some literature explores various techniques, and approaches for addressing concept drift in credit card fraud detection (Tama *et al*, 2014; Rahman *et al*, 2019; Khan *et al*, 2017; Gajbhiye and Sanyal, 2015). These papers discuss various detection methods, feature selection strategies, adaptive learning algorithms, and evaluation metrics used to handle concept drift in this domain.

4.1 Challenges and Implications

Fraudsters continually adapt their strategies to bypass fraud detection systems. They may change their tactics, such as altering transaction amounts, modifying the timing of transactions, or employing new techniques to obfuscate their activities. These changes can lead to a shift in the data distribution and make existing fraud detection models less effective. Credit card fraud detection datasets are often highly imbalanced, with a large majority of transactions being legitimate (non-fraudulent) and only a small fraction being fraudulent. Concept drift can further exacerbate this imbalance, as the distribution of fraudulent transactions may change over time. This imbalance can make it challenging to train and update models effectively.

It's important to differentiate between data drift and concept drift. Data drift refers to changes in the statistical properties of the input data, such as changes in transaction volumes, customer behaviour, or other contextual factors. Concept drift, on the other hand, pertains to changes in the underlying concept or patterns being learned by the model. Concept drift often results from data drift, but not all data drift leads to concept drift. Concept drift can also impact the relevance and importance of features used for fraud detection. Some features that were initially informative in detecting fraud may become less relevant over time, while new features may become more important. Regular feature analysis and selection techniques can help identify and adapt to feature drift caused by concept drift.

4.2 Approaches to Handle Concept Drift

Monitoring and Adaptation: To address concept drift in credit card fraud detection, continuous monitoring of the system's performance is crucial. Real-time monitoring allows for the identification of changes in the data distribution and the detection of potential concept drift. When concept drift is detected, the models and detection algorithms should be adapted or updated to accommodate the new patterns and characteristics of fraudulent transactions. Incremental Learning and Online Training: Traditional batch training approaches may not be sufficient to handle concept drift. Incremental learning techniques, such as online learning, can enable models to adapt to new data and learn from it in real-time. Online training allows the model to incorporate new instances and update its parameters as new fraud patterns emerge. Ensemble learning techniques, such as ensemble of classifiers or ensemble of models, can help mitigate the impact of concept drift. By combining predictions from multiple models trained on different subsets of data or using different algorithms, ensembles can provide more robust and adaptive fraud detection performance.

4.3 Review on Concept Drift in Credit Card Fraud Detection

"Concept Drift Detection and Model Adaptation for Credit Card Fraud Detection" was presented by Widodo and Yang (2015). This paper proposed a concept drift detection method based on the Kolmogorov-Smirnov test and presents an adaptive fraud detection model that adjusts to changing patterns. In Bhattacharyya *et al*. (2019), "Credit Card Fraud Detection Using Machine Learning: A Review" was presented. This review article discussed the challenges of concept drift in credit card fraud detection and provides an overview of various machine learning techniques and approaches used to address this problem. "A Study of Concept Drift in Credit Card Fraud Detection" was done by Tama *et al*. (2018). The authors analysed the impact of concept drift on credit card fraud detection and propose a framework that combines online learning and ensemble techniques to adapt to changing fraud patterns. "Detecting Concept Drift in Credit Card Fraud Detection Systems" was presented by Martens and Provost (2014). This study explores the detection of concept drift in credit card fraud detection systems and compares different drift detection algorithms, such as the Page-Hinkley test and the Drift Detection Method. "Adaptive Fraud Detection for Evolving Data Streams" by Lazarevic *et al*. (2003): The authors proposed an adaptive approach to fraud detection in evolving data streams, focusing on the problem of concept drift and the need for continuous model updates.

5.0 Review of Related Literature on Machine Learning Techniques for Credit Card Fraud Detection

A comprehensive survey of data mining-based fraud detection was conducted in Phua *et al* (2010). The paper categorises, compares, and summarises from almost all published technical and review articles in automated fraud detection within the last 10 years. It defines the professional fraudster, formalises the main types and subtypes of known fraud, and presents the nature of data evidence collected within affected industries. In “Learned lessons in credit card fraud detection from a practitioner perspective,” Pozzolo *et al* (2015) provided some answers from a practitioner’s perspective by focusing on three crucial issues: unbalancedness, non-stationarity and assessment. The analysis was made possible by a real credit card dataset provided by an industrial partner. Pozzolo *et al* (2015) presented a paper a realistic modelling and a novel learning strategy for credit card fraud detection. The work identified the problem which involves several relevant challenges, namely: concept drift (customers' habits evolve and fraudsters change their strategies over time), class imbalance (genuine transactions far outnumber frauds), and verification latency (only a small set of transactions are timely checked by investigators). The paper stated that many learning algorithms that have been proposed for fraud detection rely on assumptions that hardly hold in a real-world fraud-detection system (FDS). This lack of realism concerns two main aspects: 1) the way and timing with which supervised information is provided and 2) the measures used to assess fraud-detection performance. The paper made three major contributions. First, proposed a formalization of the fraud-detection problem that realistically describes the operating conditions of FDSs that everyday analyse massive streams of credit card transactions. Second, designed and assessed a novel learning strategy that effectively addresses class imbalance, concept drift, and verification latency. Third, in the experiments, demonstrated the impact of class unbalance and concept drift in a real-world data stream containing more than 75 million transactions.

Credit card fraud detection using convolutional neural networks was carried out in Li *et al* (2017). Conventional methods use rule-based expert systems to detect fraud behaviours, neglecting diverse situations, extreme imbalance of positive and negative samples. In this paper, the authors proposed a CNN-based fraud detection framework, to capture the intrinsic patterns of fraud behaviours learned from labelled data. Abundant transaction data is represented by a feature matrix, on which a convolutional neural network is applied to identify a set of latent patterns for each sample. Experiments on real-world massive transactions of a major commercial bank demonstrate its superior performance compared with some state-of-the-art methods.

In this survey (Bhattacharya *et al*, 2018) data-driven credit card fraud detection particularities and several machine learning methods were explored to address each of its intricate challenges with the goal to identify fraudulent transactions that have been issued illegitimately on behalf of the rightful card owner. The paper characterized a typical credit card detection task: the dataset and its attributes, the metric choice along with some methods to handle such unbalanced datasets. Then focussed on dataset concept drift, which refers to the fact that the underlying distribution generating the dataset evolves over times: Afterwards it highlighted different approaches used to capture the sequential properties of credit card transactions. These approaches range from feature engineering techniques to proper sequence modelling methods such as recurrent neural networks (LSTM) or graphical models (Hidden Markov models). Wang *et al.*, (2018) designed "a deep auto-encoder" by using the similarity of vector representations. To recreate unique bipartite chart taxonomy and estimated that experimental comparability about user conduct. That profound embedding comes about Might after that be utilized for compelling fraud block identification by position circulation of client vector representations. DeepFD not just altogether outperforms other benchmark methods but will be likewise All the stronger for programmed identification of various duplicity obstructs. This paper (Xuan *et al.*, 2018), examined the performance of two types of random forest models. The primary is the ordinary random forest, during every interior node, it haphazardly selects a subset of qualities. Furthermore, it computes those focuses on different classes. The second algorithm is CART (Classification Also Regression Trees), during each node, it parts dataset. Toward picking those best quality done a subset of qualities as stated by Gini impurity which measures vulnerability about the dataset. The first algorithm gives Accuracy (91.96%), Precision (90.27%), Recall (67.89%) and F-Measure (0.7811), while the second algorithm gives Accuracy (96.77%), Precision (89.46%), Recall (95.27%) and F-Measure (0.9601). Roy *et al.*, (2018), evaluates the general artificial neural network (as a part of Deep Learning topologies) with built-in topologies like memory and time components such as Long Short-term memory – and changing parameters with respect to their effectiveness in fraud detection. In Yee *et al*., (2018), this paper using Bayesian network classifiers namely Tree Augmented Naïve Bayes (TAN), K2, logistics, J48, and Naïve Bayes classifiers which is all supervised based was presented. All the classifiers give more than 95.0% accuracy when pre-processing the dataset using Principal Component Analysis and normalization, in contrast to scores obtained before pre-processing the dataset. In general, all the Bayesian classifiers give notably better scores after being fed with filtered data. In (Shakya, 2018), the paper uses a combination of different resampling techniques with Predictive models such as random forest, logistic regression, and XGBoost to prophesy if a transaction is legitimate or fraudulent. A hybrid resampling approach of Tomek Links removal and Synthetic Minority Over-Sampling Technique (SMOTE) with random forest algorithm giving best results coppered with other models.

Pham *et al* (2019) presented Deep learning for credit card fraud detection. Some major challenges identified in this paper in solving credit card frauds involve the availability of public data, high class imbalance in data, changing nature of frauds and the high number of false alarms. This paper presents a thorough study of deep learning methods for the credit card fraud detection problem and compare their performance with various machine learning algorithms on three different financial datasets. Experimental results showed great performance of the proposed deep learning methods against traditional machine learning models and implied that the proposed approaches can be implemented effectively for real-world credit card fraud detection systems. A state-of-the-art survey on deep learning theory and architectures was presented (Alom *et al*, 2019). The study presented a brief survey on the advances that have occurred in Deep Learning (DL), starting with the Deep Neural Network (DNN). The survey went on to cover Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), including Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU), Auto-Encoder (AE), Deep Belief Network (DBN), Generative Adversarial Network (GAN), and Deep Reinforcement Learning (DRL) in solving credit card fraud detection. Saputra and Suharjito, (2019) presented Fraud detection using machine learning in e-commerce. This study processes the imbalances in the datasets of e-commerce fraud transaction by using the Synthetic Minority over Sampling Technique (SMOTE) and approved that the SMOT increasing the performance of data classification, which is always unbalanced data, then used the neural network, Naïve Bayes, random forest and decision tree. The results proved that the highest accuracy was for ANN with 96%, then RF and NB were 95%, for DT accuracy was 91. Zhang *et al*., (2019) presented a paper on "HOBA: A novel feature engineering methodology for credit card fraud detection with a deep learning architecture." Their fraud detection system was based on homogeneity-oriented behaviour analysis (HOBA) which combines an advanced feature engineering process with a deep learning architecture. They approved the efficiency of their model by comparing it with other models like benchmark methods. In this work (Chen *et al*., 2019), they present a fraud detection system based on a graph- for large-scale e-commerce insurance with the cases of the most popular insurance – the security deposit insurance and the return-freight insurance". They also identify the modules and their functionality in this system.

A hybrid method for credit card fraud detection using machine learning algorithm was presented (Ramyashree *et al*., 2019). A "machine learning algorithm" produced as one of the standard models, which is used to detect fraud in e-commerce transactions. After that, add, "AdaBoost and majority vote method", which is a hybrid method. In addition, this powerful algorithm can perform an evaluation of the noisy data samples. Good accuracy rates provided

by "majority voting". 0.942 for 30% added noise is the score of several voting methods. Then, it was consummated that the voting method indicated a lot of stable execution in the presence of noise. In this paper (Lucas *et al*., 2019), they benefit from the temporal data to produce "a strategy to quantify the covariate shift." This strategy classifies the transactions in each day versus every other (that is there is a covariate shift between days when the classification is good and similar if the classification is not efficient. It is used as a distance matrix describing the covariate shift between days. Then applying the agglomerative clustering algorithm on it. Using a Random Forest classifier, they show that coordinating the information of the sort of the day previously recognized increments the Recall Precision-AUC by (2.5%). D.G *et al*., (2019) proposed a hybrid method called " CPNN-GA." The model consists of a combination of a genetic and artificial neural network algorithms for the anomaly detection in online transaction systems, where GA is an efficient algorithm for selecting optimal parameter so that optimize the quality of solution regarding classification with rate low false alarm rate and high fraud catching. Experiments show that the accuracy of CPPN is (84.42), GA (89.42) but CPPN-GA (95.58 9) with the highest hit rate (97.19). In this study (Benchaji *et al*., 2019), the writers proposed an oversampling strategy by using K-Means clustering and genetic algorithm to increase the enhancement of fraud detection in e-banking. This strategy used in imbalanced dataset's minority class for data. The model contributed to the reduction of the number of false alerts and fraudulent transactions.
The ensemble approach proposed in this paper (Carta *et al*., 2019) can deal with some familiar problems that influence classification problems by specifies this problem and the volume of risks to achieve higher precision in fraudulent transactions compared with other classification approaches. This model dealing with problems like data imbalance and increasing the number of frauds which detect correctly. Yang *et al*., (2019) solved the problem of decreasing classification performance which comes from the noisy points in AdaBoost, resulting, via clustering algorithm. The strategy of this algorithm is to determine noisy points in the group of iterations. And in each iteration for every cluster, it computes a misclassification degree which is used to determine if a misclassified sample in the current iteration is a noisy point or not. Moreover, they produced an elastic approach for the misclassified samples to update the weights. Their method achieved better results than the state-of-the-art methods including LogitBoost, AdaBoost, AdaCoast, and SPLBoost. In (Varmedja *et al*., 2019), their study was about comparison among some machine learning algorithms. The comparison confirmed that the Random Forest algorithm gives the best classification of whether transactions are fraud or not (RF with accuracy (99.96%), LR (97.46 %.) and NB (99.23%). Xie *et al*., (2019) proposed a novel approach of feature engineering for credit card fraud detection, by generate group features from individual behaviour. The problem of temporal features exceeded effectively. Their method approved its effectiveness in improving the performance of fraud detection. Simi. M, (2019) carried out comparative analysis among (Random Forest, SVM, and ANN) machine learning algorithms for credit card fraud detection. They concluded that is Random Forest has more accuracy than SVM and less accuracy than ANN in fraud detection. Porwal and Mukund (2019) presented an ensemble of clustering methods in large data sets to detect fraudulent patterns by using a consistency score for each data point. Their method approved their power in outlier detection in large datasets and changing patterns. Sadgali *et al*., (2019) reviewed the methods and techniques in "Performance of machine learning techniques in the detection of financial frauds" which showed superior results. They stated that hybrid fraud detection techniques are

the most frequently used in fraud detection methods by determining the methods and techniques that give the best outcomes that have been idealized up until this point.

In this work (Devi *et al*., 2019), a random forest algorithm was developed with cost-sensitive weights to improve the effectiveness of credit card fraud detection. In the training phase and for each tree, a cost-function has been defined, more weight has been assigned for the minority instances during training. The trees are arranged as stated by their ability prediction of the minority class instances. The model has not been checked for high-dimensional datasets. The performance of the proposed method is (F-measure [76.815], G-mean [82.298], and AUC [0.778]. In this paper (Makki *et al*., 2019), they make a comparison among eight machine learning algorithms performance, and the class imbalance problem was solved by some solutions. They studied the weakness and the performance of approaches applied to credit card fraud detection. They found that the LR, C5.0 decision tree algorithm, SVM, and ANN are the best methods according to the 3 considered performance measures (Accuracy, Sensitivity, and AUPRC). In (Shirgave *et al*., 2019), various machine learning algorithm were reviewed for fraud detection in credit card transaction with the performance based on precision, accuracy, and specificity metrics. The alert has been checked as fraudulent or authorized by using the Random Forest classifier. By using delayed and feedback supervised sample. The classifier gave an accuracy (0.962), Precision (0.997), and Specificity (0.987).

Li *et al* (2020) presented a survey on Fraud detection in e-commerce. Lee *et al*, (2020) presented a systematic review study on Machine learning for credit card fraud detection while, Bhattacharya and Zhang (2021) presented a survey on fraud detection techniques in electronic payment systems. Nakai (2020) developed the fraud detection model without a supervised label based on anomaly detection and succeeded in detecting certified fraud contracts in the top rank. He solved the problem of the impossibility to detect fraud in the supervised label when the number of fraudulent transactions is extremely few. In this paper (Lucas *et al*., 2020), a feature engineering strategy has been adopted by HMM, which permits them to integrate consecutive knowledge in the transactions in the form of HMM-based features. These features which based on HMM- could make a Random Forest classifier (non-sequential classifier) to use sequential information for the classification process. The strategy of the automated feature engineering based on HMM has "multiple perspective property" which enabled them to benefit from incorporating a wide range of sequential information. The study differentiates the genuine from fraudulent behaviours of the cardholders & the merchants with respect to timing and amount features of the transactions. Alazizi *et al*., (2020), produced a new fraud detection technique called "DuSVAE model". The model is a combination of two" sequential variational auto-encoders" used in the input sequential data to construct a concentrated representation vector which then used as input to the classifier to give it the possibility of classification transactions as genuine or fraudulent. Mehana and Nuci, (2020) presented a paper on “Fraud Detection using Data-Driven approach." An incremental learning approach was introduced as a new method for real-time fraud detection in online banking transactions. The model AFDM analysed transactions based on the transaction instead of accounts which gives more detailed information and represents a good assistant next to detection actions. In addition, the knowledge of the classification algorithm like NB can increment its updatable transaction by transaction. This novel approach makes the detection and response possibility in real-time, and it gives an accuracy (97,2). It also allowed learning new concepts of behaviour changes immediately. Meng *et al*., (2020) presented “A case study in credit fraud detection with SMOTE and XGboost.” The fraudulent

credit card transactions have been detected by using a hybrid method consists of an XGBoost algorithm preceded with a SMOTE algorithm which is in turn use to achieve the balance of data. The data balancing made the model more generalized and stable.
Alghofaili *et al*., (2020) used the "Long Short-Term Memory (LSTM) technique" to improve the traditional financial fraud detection systems. The model identifies unknown and sophisticated patterns of fraud quickly and with high accuracy (99.95% in less than a minute) on a real dataset. In (Daliri, 2020) a hybrid system based on the Artificial Neural Network (ANN) technique and Harmony Search Algorithm (HSA) was used to detect fraud. The parameters of ANN have been optimized by using HSA, and ANN is used to detect fraud. The system show accuracy with (86) and recall criteria (87). An "oversampling method based on the VAE" has been proposed as a new approach for fraud credit card detection (Tingfei *et al*., 2020). This approach is a new supervised oversampling method that can deal with the class imbalance of the datasets, but it is impossible to apply to unsupervised systems. Compared with SMOTE and GAN it shows less performance in recall metric and when the model deals with novel fraud data could perform badly. Babu *et al*., (2020) evaluated the performance of "deliver regression, call tree and random woodland" to detecting the fraud in grasp card. Depending on the results, finding that selecting a random forest algorithm over XGboost can be a reasonable method to get the next degree of Inclusiveness with a little decrease in performance. Safont and Vergara (2020) presented a novel approach to increase fraud detection performance by using "the differences in temporal dependence (sequential patterns)" between valid and non-legitimate credit card operations. It is assumed that in feature spaces with low-dimension, the successive patterns are more notable than the features with high-dimension space of all the card operation records. Linear and quadratic discriminant analyses, classification tree, and naive Bayes are the single classifiers used in this model. With these single classifiers and to get better results, Alpha integration has been applied.
An improvement is proposed (Zheng *et al*., 2020) to the TrAdaBoost algorithm (ITrAdaBoost) by changes the weight of the instance, which is wrongly classified in a source scope, and then calculate the distance depending on the theory of “reproducing kernel Hilbert space". In addition, the concept drift problem in the transactions has been solved by applying the ITrAdaBoost algorithm. Unfortunately, the computation time is high because of the distribution distances. An "optimized light gradient boosting machine (OLightGBM)" has been Proposed as an intelligent method of credit card detection (Taha and Malebary, 2020). Experimental results indicate that the proposed approach exceeded the other machine learning algorithms including (RF, LR, R-SVM, linear -SVM, KNN, DT, and NB) and achieved the highest performance in terms of Accuracy (98.40%), AUC (92.88%), Precision (97.34%) and F1-score (56.95%). Mittal (2020) proposed an efficient learning strategy to determine the challenges (classification problem and performance measures) including "verification latency and audio feedback communication". This learning strategy applied to a wide range of credit card transactions. Jagdish *et al*., (2020) proposed "MFDA (Modified Fisher Discriminant Analysis)" for fraud credit card detection. The study solves the shortcoming in FDM, the generated results were not accurate when using input values with little less sensitivity. In addition to MFDA, several algorithms are used (DT, ANN, NB, and Normal Fisher). Experiments show that FDMA has the highest accuracy. A feedback system mechanism based on machine learning methodology has been introduced as efficient fraud detection for a credit card (Trivedi *et al*., 2020). This method improves the rate of classifier

detection aside from cost-effectiveness. The experiments show that random forest techniques show an accuracy percentage of 95.988 %, although SVM 93.228 %, LR 92.89 %, NB 91.2 %, Decision trees 90.9 %. Priscilla and Prabha (2020) presented a review on Credit Card Fraud Detection (CCFD), in solving the problem of class imbalance in datasets and Machine Learning approaches. The conclusion of the review is Supervised Learning is highly used than the unsupervised for CCFD, the fraud detection techniques which are most used are SVM, Bayesian classifiers, LR, and DT, and Random Forest is the most usage frequency Ensemble learning technique with a good performance. finally, Hybrid methods are better than using a single classifier. Li *et al*., (2021) improved the support vector machine parameters by using a cuckoo search (CS) algorithm to raise the ability to detect fraud of credit cards. The results demonstrate that CS-SVM is superior to SVM in Accuracy, Precision, Recall, F1-score, AUC, and superior to LR, RF, DT, and NB, whose accuracy is 98%.

6.0 Comparative Analysis of Credit Card Fraud Detection Methods and Performance

Evaluating and comparing the performance of different machine learning techniques in credit card fraud detection is crucial to determining the most effective approach. Performance evaluation involves assessing the algorithms based on various metrics such as accuracy, precision, recall, F1 score, and area under the curve (AUC). Numerous studies have explored this topic, considering various algorithms and evaluation metrics. Below is an overview of the performance evaluation of machine learning techniques in credit card fraud detection, along with some notable evaluation metrics and studies:

1. Accuracy: Accuracy measures the overall correctness of the predictions made by a model. However, in the case of imbalanced datasets where most transactions are non-fraudulent, accuracy alone can be misleading. Bhattacharyya *et al*. (2019) and Dal Pozzolo *et al*. (2015) conducted studies that included accuracy as one of the evaluation metrics for comparing different machine learning techniques in credit card fraud detection.
2. Precision and Recall: Precision measures the proportion of correctly identified fraudulent transactions out of the total transactions classified as fraudulent. Recall (also known as sensitivity) measures the proportion of correctly identified fraudulent transactions out of the actual fraudulent transactions. Studies by Kwon and Kim (2019), Ribeiro *et al*. (2016), and Sathya and Padma (2018) evaluated machine learning techniques using precision and recall assessing their performance in credit card fraud detection.
3. F1 Score: The F1 score is the harmonic mean of precision and recall, providing a balanced evaluation metric that considers both false positives and false negatives. Several studies, including those by Bhattacharyya *et al*. (2019), Ribeiro *et al*. (2016), and Sathya and Padma (2018), utilized the F1 score to compare the performance of machine learning techniques in credit card fraud detection.
4. Area Under the Curve (AUC): The AUC is a popular metric used in evaluating the performance of binary classification models. It represents the trade-off between true positive rate (sensitivity) and false positive rate. Some studies, such as Phua *et al*. (2010), utilized AUC to assess the performance of machine learning techniques in credit card fraud detection.
5. Receiver Operating Characteristic (ROC) Curve: The ROC curve is a graphical representation of the true positive rate against the false positive rate at various classification thresholds. It is often used in conjunction with AUC to evaluate the performance of machine

learning techniques. Studies by Phua *et al*. (2010) and Ribeiro *et al*. (2016) utilized ROC curves to compare the performance of different machine learning algorithms in credit card fraud detection.

It is important to note that the performance of machine learning techniques in credit card fraud detection can vary depending on the dataset, feature engineering, model configuration, and evaluation metrics used. Furthermore, the performance comparison may differ in different studies due to variations in the dataset and experimental setup. Hence, it is recommended to conduct comparative evaluations using appropriate evaluation metrics on specific datasets to identify the most suitable machine learning technique for credit card fraud detection in each context.

In most literature the main metrics used in evaluating the result of fraud detection systems include Precision, Accuracy and Recall. Hence, different methods give different evaluation depending on the method or methods used in the system. Various techniques in machine learning used in solving credit card fraud detection problem fall under either supervised, unsupervised, ensemble, and deep learning. Table shows an updated comparative study of machine learning techniques used in solving credit card fraud detection (Priscilla and Prabha, 2020). Some of the reviewed literature were not explicit in reporting their evaluation metrics with other models using Precision, Recall and/or Accuracy of prediction.

Table 1: Comparative Analysis of Machine Learning Techniques used in Credit Card Fraud Detection

Machine learning	*Technique*	*Accuracy*	*Precision*	*Recall*	*References*	*Pros*	*Cons*
Supervised Learning	*Artificial Neural Network (ANN)*	*98.69*	*98.41*	*98.98*	*[4,17,19,20,32,36,38,52,53,54,55,56,57,72,75,78,82]*	*Highly accurate and reliable*	*Need to understand and label the input. More computation time required for the training stages.*
	Support Vector Machine (SVM)	*93.96*	*93.22*	*93.0*	*[21,33,38,39,53,59,61,62,63,64,65,66,67,68,69,70,71,74,75,78]*		
	Naïve Bayes Classifier (NB)	*91.62*	*97.09*	*84.82*	*[20,21,35,39,42,47,55,56,58,59,61,64,66,67,73,74,78,80,82,87,93]*		
	k-Nearest Neighbour (kNN)	*94.99*	*94.58*	*92.00*	*[39,64,67,73,81,82]*		
	Logistic Regression (LR)	*94.84*	*97.58*	*92.00*	*[21,33,36,39,42,52,53,55,59,61,63,64,66,71,73,74,75,82,83,84,85]*		
	Decision Trees (DT)	*92.88*	*99.48*	*86.34*	*[20,21,33,35,39,51,52,53,55,59,63,64,66,69,73,74,75,82,83,84,85]*		

Unsupervised Learning	*Expectation-Maximization*	*83.70*			*[58]*	*Easy to find unknown patterns within of data. Useful when there are limited number of labelled data.*	*High Computation time. Less accurate due to unlabelled input.*
	K-Means Clustering	*72.96*			*[14,58,64]*		
	FuzzyC-Means	*93.90*			*[57]*		
	DBSCAN				*[55]*		
	Hidden Markov Model (HMM)	*98.00*			*[23,55,56,8 9]*		
	Self-Organizing Map (SOM)				*[55]*		
Ensemble Learning	*Random Forests (RF)*	*99.96*	*96.38*	*81.63*	*[13,18,21,22,23,35,36, 37,38,39,42,45,51,52,53,54,59,61, 63,65,66,67,69,71,72,78,81,82,84, 85,91,93]*	*Avoids the overfitting problem and gives better predictions when compared with a single model classifier.*	*Computation time is high. Reduces model interpretability due to increased complexity.*
	Boosting	*86.11*	*90.90*		*[53,65,69,70,71,82,92]*		
	Bagging	*96.00*			*[64,93,64]*		
	Voting	*96.00*			*[93]*		
Deep Learning	*Stochastic Gradient Descent*	*84.70*			*[82]*	*No need for feature extraction and labelling of data*	*A large amount of data is needed to find the pattern. Create overfitting problem in the model.*
	Long short-term Memory	*85.00*			*[32,26,61, 91]*		
	Deep Feed Forward Neural Network				*[83,92,95]*		
	Variational Autoencoder (VAE)	*99.96*			*[10,40,95]*		
	Auto Encoder (AE)	*99.00*			*[68,54]*		
	Restricted Boltzmann Machines				*[54]*		
	Recurrent Neural Network				*[32,61]*		
	Convolutional Neural Network (CNN)	*85.00*			*[54]*		
	Generative Adversarial Network (GAN)	*99.96*			*[40, 98]*		

Most fraud detection systems use supervised learning while the ensemble techniques show better performance. Furthermore, many studies reveal that the hybrid methods show higher performance than using one method and using search algorithms to optimize parameters of classification algorithms give the best results.

7.0 Challenges and Future Directions

Credit card fraud detection using machine learning and concept drift techniques present several challenges and offers exciting future directions for research and development. Here are some key challenges and potential future directions in this field.

Credit card fraud datasets are highly imbalanced, with a small number of fraud instances compared to legitimate transactions. Addressing this class imbalance is essential for effectively training machine learning models. Fraud patterns evolve over time, leading to concept drift. Adapting to changing fraud patterns and detecting concept drift in real-time is a significant challenge in credit card fraud detection.

Selecting relevant features and engineering them appropriately is crucial for effective fraud detection. Incorporating domain knowledge and developing innovative feature engineering techniques can enhance model performance. Timeliness is critical in fraud detection to prevent financial losses. Developing real-time detection systems that can analyze transactions and make accurate fraud decisions within milliseconds is challenging yet essential. There should be more research in real-time detection systems.

Machine learning models used in fraud detection often lack transparency and interpretability. Developing techniques to explain and interpret the decisions made by these models is important for building trust and understanding the factors influencing fraud detection. Adversarial Attacks: Fraudsters continuously adapt and devise strategies to evade detection systems. Developing robust models that are resilient to adversarial attacks, such as data poisoning or evasion attacks, is a challenge that needs to be addressed.

Future areas of research should be geared towards advanced machine learning techniques such as deep learning models, reinforcement learning to improve credit card fraud detection performance. Investigating ensemble learning methods, such as stacking, boosting, or bagging, to combine multiple models and enhance fraud detection accuracy can be a promising direction for future research. Developing algorithms that can adapt to concept drift in real-time by utilizing online learning or incremental learning techniques can improve the effectiveness of fraud detection systems. Incorporating multiple data sources, such as transaction data, social network data, or user behaviour data, and fusing them to provide a comprehensive view of fraudulent activities can enhance fraud detection capabilities. Ensuring privacy and compliance with data protection regulations is crucial. Exploring privacy-preserving techniques, such as federated learning or differential privacy, can enable effective fraud detection while safeguarding sensitive user information. Creating benchmark datasets with standardized evaluation metrics will facilitate fair comparisons between different fraud detection techniques and promote reproducibility of research results. Finally, collaborative efforts among researchers, financial institutions, and regulatory bodies to share anonymized data can drive advancements in credit card fraud detection research and enable the development of more effective models.

In conclusion, credit card fraud detection using machine learning and concept drift techniques faces challenges related to imbalanced data, concept drift, feature engineering, and model interpretability. Future directions include exploring advanced machine learning techniques, ensemble learning, online learning, multi-modal data fusion, privacy preservation, standardized evaluation metrics, and collaborative efforts. Addressing these challenges and exploring the directions can lead to more accurate and robust credit card fraud detection systems.

8.0 Conclusion

This study highlights the importance of machine learning techniques and concept drift handling methods for credit card fraud detection. Various studies have compared different machine learning algorithms and concept drift handling techniques, showing that ensemble methods, neural networks, and adaptive models perform well in detecting fraudulent transactions. Additionally, the use of hybrid approaches and ensemble techniques has shown promising results in adapting to concept drift. Future research should focus on addressing challenges such as data imbalance, real-time fraud detection, and interpretability of models to further improve credit card fraud detection.

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